Solution of least squares problem

- we make one assumption: A *has linearly independent columns*
- this implies that Gram matrix $A^T A$ is invertible
- unique solution of least squares problem is

$$\hat{x} = (A^T A)^{-1} A^T b = A^\dagger b$$

- cf. $x = A^{-1} b$, solution of square invertible system $Ax = b$

Derivation via calculus

- define

$$f(x) = \|Ax - b\|^2 = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} A_{ij} x_j - b_i \right)^2$$

- solution $\hat{x}$ satisfies

$$\frac{\partial f}{\partial x_k}(\hat{x}) = \nabla f(\hat{x})_k = 0, \quad k = 1, \ldots, n$$

- taking partial derivatives we get $\nabla f(x)_k = \left(2A^T(Ax - b)\right)_k$
- in matrix-vector notation: $\nabla f(\hat{x}) = 2A^T(A\hat{x} - b) = 0$
- so $\hat{x}$ satisfies *normal equations* $(A^T A)\hat{x} = A^T b$
- and therefore $\hat{x} = (A^T A)^{-1} A^T b$

Direct verification

- let $\hat{x} = (A^T A)^{-1} A^T b$, so $A^T(A\hat{x} - b) = 0$
- for any n-vector x we have

$$\begin{aligned} \|Ax - b\|^2 &= \|(Ax - A\hat{x}) + (A\hat{x} - b)\|^2 \\ &= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 + 2(A(x - \hat{x}))^T(A\hat{x} - b) \\ &= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 + 2(x - \hat{x})^T A^T(A\hat{x} - b) \\ &= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 \end{aligned}$$

- so for any x, $\|Ax - b\|^2 \geq \|A\hat{x} - b\|^2$
- if equality holds, $A(x - \hat{x}) = 0$, which implies $x = \hat{x}$ since columns of A are linearly independent

Computing least squares approximate solutions

- compute QR factorization of A: $A = QR$ $\quad (2mn^2$ flops)
- QR factorization exists since columns of A are linearly independent
- to compute $\hat{x} = A^\dagger b = R^{-1}Q^T b$
  - form $Q^T b$ $\quad (2mn$ flops)
  - compute $\hat{x} = R^{-1}(Q^T b)$ via back substitution $\quad (n^2$ flops)
- total complexity $2mn^2$ flops

- identical to algorithm for solving $Ax = b$ for square invertible A
- but when A is tall, gives least squares approximate solution

Outline

Least squares problem

Solution of least squares problem

Examples

Illumination

- n lamps illuminate an area divided in m regions
- A_{ij} is illumination in region i if lamp j is on with power 1, other lamps are off
- x_j is power of lamp j
- $(Ax)_i$ is illumination level at region i
- b_i is target illumination level at region i

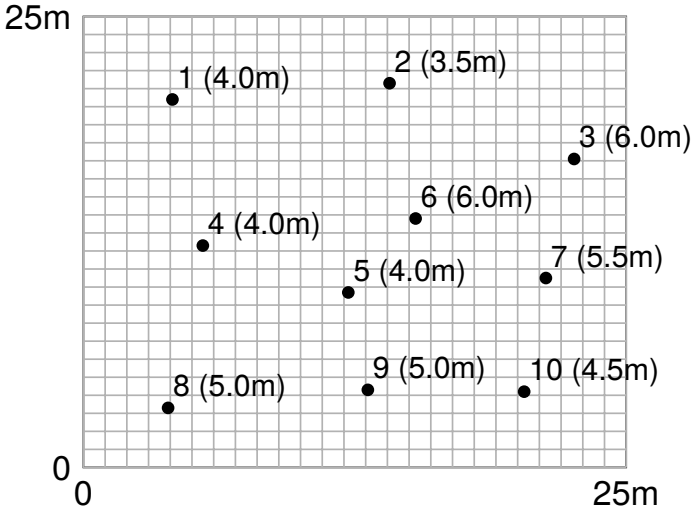
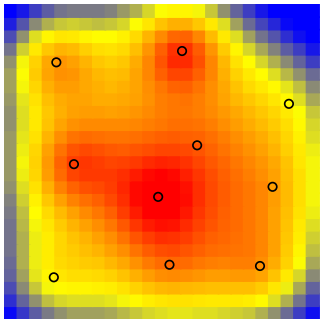


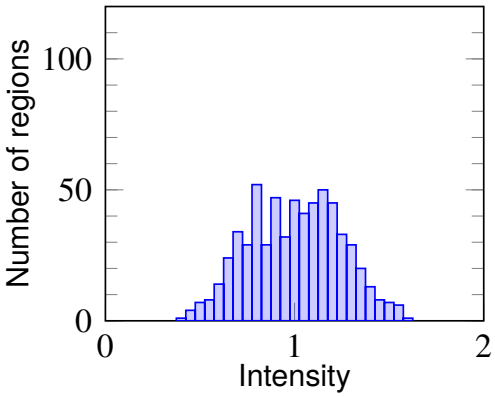
figure shows lamp positions for example with

$$m = 25^2, \quad n = 10$$

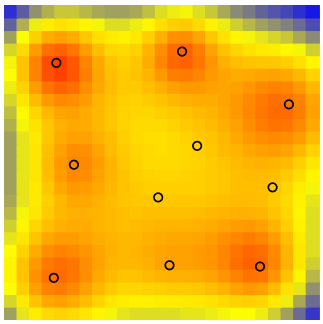
Illumination

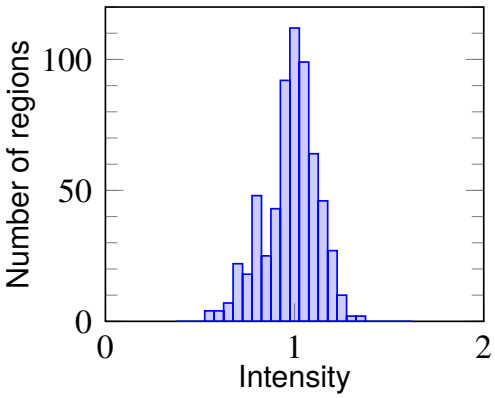
- equal lamp powers ($x = \mathbf{1}$)





- least squares solution $\hat{x}$, with $b = \mathbf{1}$





13. Least squares data fitting

Outline

Least squares model fitting

Validation

Feature engineering

Setup

- we believe a scalar y and an n-vector x are related by *model*

$$y \approx f(x)$$

- x is called the *independent variable*
- y is called the *outcome* or *response variable*
- $f : \mathbf{R}^n \to \mathbf{R}$ gives the relation between x and y
- often x is a feature vector, and y is something we want to predict
- we don't know f, which gives the 'true' relationship between x and y

Data

- we are given some *data*

$$x^{(1)}, \ldots, x^{(N)}, \qquad y^{(1)}, \ldots, y^{(N)}$$

  also called *observations*, *examples*, *samples*, or *measurements*

- $x^{(i)}, y^{(i)}$ is ith *data pair*
- $x_j^{(i)}$ is the jth component of ith data point $x^{(i)}$

Model

- choose *model* $\hat{f} : \mathbf{R}^n \to \mathbf{R}$, a *guess* or *approximation* of f
- *linear in the parameters* model form:

$$\hat{f}(x) = \theta_1 f_1(x) + \cdots + \theta_p f_p(x)$$

- $f_i : \mathbf{R}^n \to \mathbf{R}$ are *basis functions* that we choose
- θ_i are *model parameters* that we choose
- $\hat{y}^{(i)} = \hat{f}(x^{(i)})$ is (the model's) *prediction* of $y^{(i)}$
- we'd like $\hat{y}^{(i)} \approx y^{(i)}$, *i.e.*, model is consistent with observed data

Least squares data fitting

- *prediction error* or *residual* is $r^{(i)} = y^{(i)} - \hat{y}^{(i)}$
- *least squares data fitting*: choose model parameters θ_i to minimize RMS prediction error on data set

$$\left(\frac{(r^{(1)})^2 + \cdots + (r^{(N)})^2}{N} \right)^{1/2}$$

- this can be formulated (and solved) as a least squares problem

Fitting univariate functions

- when $n = 1$, we seek to approximate a function $f : \mathbf{R} \to \mathbf{R}$
- we can plot the data (x_i, y_i) and the model function $\hat{y} = \hat{f}(x)$

Straight-line fit

- $p = 2$, with $f_1(x) = 1$, $f_2(x) = x$
- model has form $\hat{f}(x) = \theta_1 + \theta_2 x$
- matrix A has form

$$A = \begin{bmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ \vdots & \vdots \\ 1 & x^{(N)} \end{bmatrix}$$

- can work out $\hat{\theta}_1$ and $\hat{\theta}_2$ explicitly:

$$\hat{f}(x) = \mathbf{avg}(y^{\mathrm{d}}) + \rho \frac{\mathbf{std}(y^{\mathrm{d}})}{\mathbf{std}(x^{\mathrm{d}})}(x - \mathbf{avg}(x^{\mathrm{d}}))$$

  where $x^{\mathrm{d}} = (x^{(1)}, \ldots, x^{(N)})$

Example

